



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 • STANDARD 6.GR.1

Determine the Area of Triangles

Lesson 5-2 • Teacher Copy

Name: _____

Date: _____

Class: _____

SKY HARBOR MISSION

The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can find the area of a triangle using the formula $A = \frac{1}{2} \times \text{base} \times \text{height}$.

Language Objective: I can explain my steps using the words base, height, area, and perpendicular.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Area of a Triangle

Base

Height

Area

Perpendicular

Composite figure

Formula

Parallel



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The space inside a triangle, found with $A = \frac{1}{2} \times b \times h$, is the

2

The side of a triangle you measure the height from at a 90° angle is the

3

The perpendicular distance from the base to the opposite vertex is the

4

The amount of flat space inside a triangle is its .

5

Two lines that meet at a 90° angle are .

6 A shape made of two or more basic shapes combined is a

.

7 A math rule written with symbols, like $A = \frac{1}{2} \times b \times h$, is a

.

8 Two sides that stay the same distance apart and never meet are

.

Write About the Math

How many cans of paint does the homeowner need?

¿Cuántas latas de pintura necesita el dueño de la casa?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Area of a Triangle**
Área de un triángulo

☐ **Base**
Base

☐ **Height**
Altura

☐ **Area**
Área

☐ **Perpendicular**
Perpendicular

Model: *The height is the perpendicular distance from the base to the top corner.* — Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

• **The wall's area is** ____ **sq ft because** $A = 1/2 \times$ ____ $\times$ ____ **. She needs** ____ **cans because** ____ $\div 50 =$ ____.

• **My answer is** ____ **because** ____.

Mi respuesta es ____ *porque* ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

• **My claim is** ____.

Mi afirmación es ____.

• **My evidence is** ____, **so** ____.

Mi evidencia es ____, *entonces* ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Area of a Triangle
2. **Notes 2:** Base
3. **Notes 3:** Height
4. **Notes 4:** Area
5. **Notes 5:** Perpendicular
6. **Notes 6:** Composite figure
7. **Notes 7:** Formula
8. **Notes 8:** Parallel
9. **Watch for this mistake:** *A common mistake in Area of Triangles is forgetting the $\frac{1}{2}$ — students multiply base $\times$ height like a rectangle and stop there, or they use the triangle's slanted side instead of the perpendicular height that's already given. Always check: did I take half, and did I use the straight-up height, not the slanted side?*
10. **Try It 1:** A. 14 sq ft — $A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 7 \times 4 = \frac{1}{2} \times 28 = 14 \text{ sq ft}$. Multiply base $\times$ height first (28), then take half (14).
11. **Try It 2:** A. 33 sq ft — $A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 11 \times 6 = \frac{1}{2} \times 66 = 33 \text{ sq ft}$.